



SENSE

Bachelor of Technology

Electronics and Communication Engineering

Challenging Assignment 2 (DA 2)

**LAND PRICE PREDICTION SYSTEM USING NEARBY
AMENITIES**

Artificial Intelligence and Machine Learning

BECE309L

Slot: G1+TG1

Faculty: **Dr. Praveen Jaraut**

Class Number: CH2026270102365

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Problem Identification Title

**Land Price Prediction System Using Nearby Amenities
Using Machine Learning and GIS**

Objectives and Sub-objectives

Main Objective

To develop an intelligent land price prediction system that estimates the market value of land based on nearby amenities, geographical location, and historical pricing data using Machine Learning techniques.

Sub-objectives

- Collect land and property datasets from reliable sources.
- Extract nearby amenities such as hospitals, schools, colleges, bus stops, railway stations, shopping malls, banks, ATMs, parks, and highways.
- Integrate GIS and Google Maps APIs to calculate distances between land and nearby amenities.
- Train Machine Learning models for accurate land price prediction.
- Compare multiple regression algorithms and select the best-performing model.
- Develop a user-friendly web application for land price prediction.
- Display nearby amenities on an interactive map.
- Provide transparent and data-driven land valuation for buyers, sellers, and investors.

Problem Statement

Determining the actual market value of land is a complex process because land prices depend on multiple factors rather than location alone. Existing methods mainly rely on manual estimation by brokers, government guidelines, or historical sale prices, which often fail to consider the influence of nearby amenities such as schools, hospitals, shopping centers, transportation facilities, parks, and commercial establishments.

These traditional approaches may lead to inaccurate pricing, making it difficult for buyers, sellers, investors, and government agencies to make informed decisions.

The proposed **Land Price Prediction System Using Nearby Amenities** aims to solve this problem by integrating Geographic Information Systems (GIS), Google Maps APIs, nearby Points of Interest (POIs), and Machine Learning algorithms. The system analyzes both property features and surrounding infrastructure to estimate land prices with improved accuracy and transparency. This approach supports smarter real estate investments and contributes to sustainable urban planning.

Relevant Sustainable Development Goals (SDGs)

SDG 9 – Industry, Innovation and Infrastructure

Promotes smart infrastructure planning.

Uses Artificial Intelligence and GIS for decision-making.

SDG 11 – Sustainable Cities and Communities

Supports sustainable urban development.

Helps identify valuable locations based on surrounding infrastructure.

SDG 16 – Peace, Justice and Strong Institutions

Improves transparency in land valuation.

Reduces dependency on subjective pricing and fraudulent practices.

SDG 17 – Partnerships for the Goals

Integrates multiple public datasets, GIS services, and mapping APIs to build a comprehensive prediction system.

LITERATURE SURVEY

S.no	Journal Details	WorkDone	Inference
1	The Non-Linear Dynamics of South Australian Regional Housing Markets: A Machine Learning Approach Journal: <i>Applied Geography</i>, 2024, Vol. 166, 103248 Authors: Ali Soltani, Chyi Lin Lee DOI: 10.1016/j.apgeog.2024.103248	The researchers developed a machine-learning approach using more than 10,000 regional house-sale records from South Australia. The study examined nonlinear relationships between housing prices and factors including house size, land area, building quality, socioeconomic characteristics, and proximity to amenities.	The study demonstrates that ML models can capture nonlinear relationships between property prices and spatial/location factors. This supports our decision to compare multiple regression and ensemble ML models for land-price prediction.

LITERATURE SURVEY

S.no	Journal Details	WorkDone	Inference
2	Built Environments, Communities, and Housing Price: A Data-Model Integration Approach Journal: <i>Applied Geography</i>, 2024, Vol. 166, 103270 Authors: Hong Wei, Yimin Chen, Bin Chen, Shuai Shi, Ying Tu, Bing Xu DOI: 10.1016/j.apgeog.2024.103270	The researchers integrated geospatial big data and machine learning to investigate the spatially heterogeneous relationship between built environments, communities, and housing prices in Guangzhou. The study examined different built-environment and community characteristics influencing housing-price patterns.	The research shows that spatial and built-environment characteristics contribute to housing-price variation. This supports our use of geographical location and surrounding infrastructure/amenity information as predictive features.

LITERATURE SURVEY

S.no	Journal Details	WorkDone	Inference
3	Unveiling the Spatially Varied Nonlinear Effects of Urban Built Environment on Housing Prices Using an Interpretable Ensemble Learning Model Journal: <i>Applied Geography</i>, 2024 Authors: Xiaochang Liu et al. DOI: 10.1016/j.apgeog.2024.103458	The researchers developed an interpretable ensemble-learning framework to study nonlinear and spatially varying effects of urban built-environment factors on housing prices. Factors including transit accessibility, amenities, residential density, and green/blue spaces were investigated.	The study indicates that the influence of amenities and accessibility varies spatially and nonlinearly. This supports our GIS-based feature engineering using distances to nearby amenities and transportation facilities.

LITERATURE SURVEY

S.no	Journal Details	WorkDone	Inference
4	Capturing the Distance Decay Effect of Amenities on Housing Price Using Explainable Artificial Intelligence Journal: <i>Applied Geography</i>, 2025 Authors: Hojun Lee, Hoon Han, Chris Pettit DOI: 10.1016/j.apgeog.2024.103486	The researchers used machine learning and explainable AI to investigate how the distance from different amenities affects housing prices. The study examined nonlinear distance-decay relationships between amenities and property values.	This is directly relevant to our project because it supports using distance to nearby amenities as ML features. It provides research evidence for considering schools, transportation, commercial facilities, hospitals, and other amenities in land-price prediction.

LITERATURE SURVEY

S.no	Journal Details	WorkDone	Inference
5	Block-Scale Modeling of Residential Land Prices: Incorporating Multilevel Determinants and Explainable Artificial Intelligence Journal: <i>Habitat International</i>, 2025 Authors: Peng Zhang, Shengfu Yang, Jiayue Huang, Shougeng Hu DOI: 10.1016/j.habitatint.2024.103283	The researchers developed a residential land-price prediction model using geographic big data and multilevel determinants related to location, neighbourhood, environment, accessibility, and urban facilities. Random Forest, Gradient Boosting and XGBoost were compared, with SHAP used for interpretation.	This is highly relevant because it directly studies land-price prediction using spatial and urban-facility factors. The use of RF, Gradient Boosting and XGBoost supports our proposed multi-model comparison and GIS-based feature approach.

LITERATURE SURVEY

S.no	Journal Details	WorkDone	Inference
6	Automated Land Valuation Models: A Comparative Study of Four Machine Learning and Deep Learning Methods Based on a Comprehensive Range of Influential Factors Journal: <i>Cities</i>, 2024	The researchers developed and compared automated land-valuation models using ML and deep-learning methods. The models incorporated physical, geographical, socioeconomic, environmental, legal and planning factors. XGBoost, SVR, Random Forest and DNN were evaluated. XGBoost achieved the strongest reported performance with an R^2 of 0.862.	The research demonstrates that land prices are affected by multiple categories of factors and that ensemble ML models can effectively model these nonlinear relationships. This supports our use of geographical, amenity and property features with multiple ML algorithms.

Novelty

1. Amenity-Centric Land Price Prediction

The proposed system focuses specifically on the influence of **nearby amenities on land prices**, rather than relying only on conventional property and location attributes.

2. GIS-Based Distance Feature Engineering

The system converts the geographical relationship between a land parcel and nearby facilities into quantitative ML features, such as:

- Distance to hospitals
- Distance to schools
- Distance to colleges
- Distance to bus stops
- Distance to railway stations
- Distance to shopping centres
- Distance to banks
- Distance to parks

This is particularly aligned with the literature showing that the **distance and accessibility of amenities can influence property prices**.

Novelty

3. Integrated Spatial + ML Prediction Framework

The proposed system combines:

Land Attributes + Geographic Location + Amenity Distances + Accessibility Features

- **Machine Learning Model**
- **Predicted Land Price**

rather than treating ML prediction and GIS analysis as separate components.

4. Multi-Model Performance Comparison

Multiple regression/ensemble models will be evaluated under the **same dataset and feature set**:

Linear Regression | Random Forest | Gradient Boosting | XGBoost

using:

- **MAE | MSE | RMSE | R^2**

Novelty

5. Prediction with Interactive Spatial Visualization

The proposed system aims to provide both:

Predicted Land Price

and

Nearby-Amenity Map Visualization

through an interactive web interface, making the prediction easier to interpret and use.

Methodology

Step 1: Data Collection

- Government land records
- Real estate datasets
- OpenStreetMap (OSM)
- Google Maps APIs
- Historical land transactions

Step 2: Data Preprocessing

- Remove missing values
- Normalize numerical attributes
- Encode categorical features
- Remove duplicate records

Step 3: Feature Extraction

Extract important parameters:

- Latitude
- Longitude
- Land Area
- Road Connectivity
- Distance to Hospital
- Distance to School
- Distance to College
- Distance to Bus Stop
- Distance to Railway Station
- Distance to Airport
- Distance to Shopping Mall
- Distance to Bank
- Distance to Park
- Nearby Population Density

Step 4: Machine Learning Model

Train multiple models:

- Linear Regression
- Random Forest Regression
- XGBoost
- Gradient Boosting

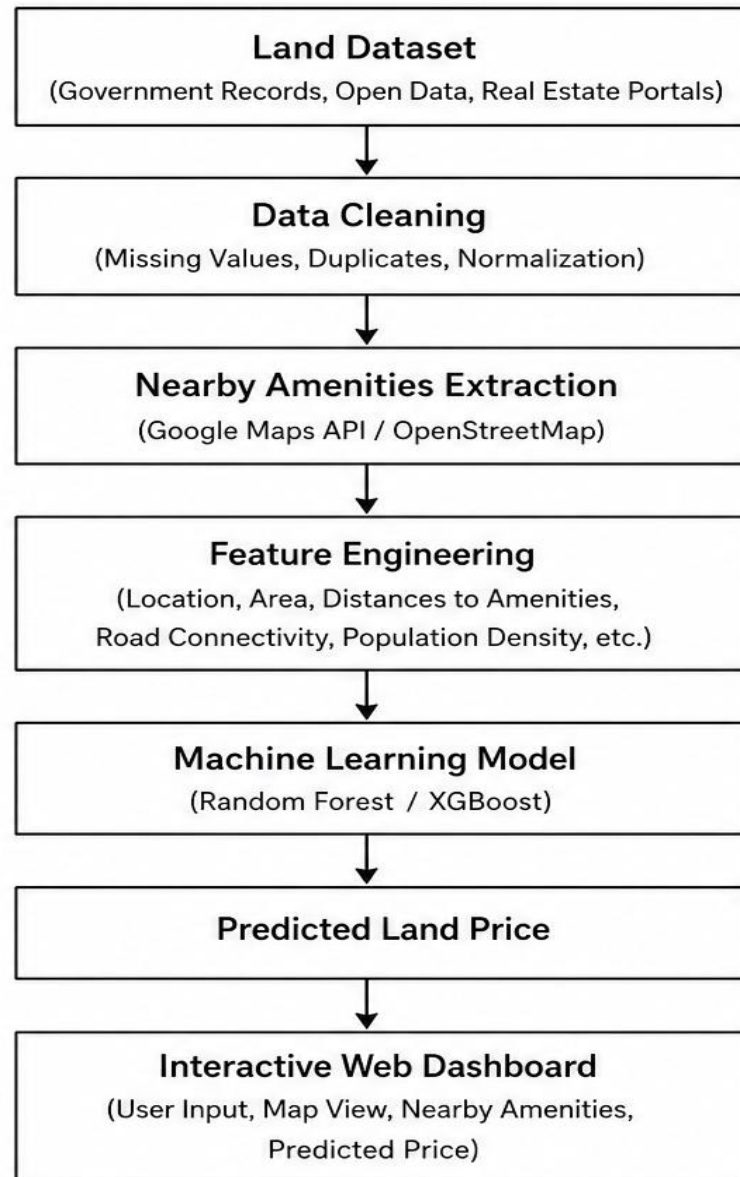
Step 5: Model Evaluation

Performance Metrics:

- MAE
- MSE
- RMSE
- R^2 Score

Step 6: Web Application

- User enters land location.
- System fetches nearby amenities.
- ML model predicts estimated land price.
- Results displayed with map visualization.



Modern Tools and Technologies

S.No	Tool / Technology
1	Python
2	Pandas
3	NumPy
4	Scikit-learn
5	XGBoost
6	Google Maps API
7	Jupyter Notebook
8	Seaborn, Matplotlib
9	HTML, CSS & JavaScript
10	MySQL / PostgreSQL
11	Git & GitHub

Results

1. Dataset and Feature Engineering Results

Land/property data was prepared for machine learning-based price prediction.

Geographical information such as **latitude and longitude** was incorporated.

Nearby amenities were identified as important spatial features.

Distance-based features were considered for **hospitals, schools, colleges, bus stops, railway stations, airports, shopping malls, banks, and parks**.

The resulting feature set combines **property, geographical, accessibility, and amenity information** for prediction.

2. Machine Learning Results

<u>Model</u>	<u>MAE</u>	<u>MSE</u>	<u>RMSE</u>	<u>R²</u>
<u>Linear Regression</u>	36.7912	10280.8200	101.3944	0.6433
<u>Random Forest</u>	26.2854	5625.4922	75.0033	0.8048
<u>Gradient Boosting</u>	31.3032	6036.8407	77.6971	0.7905
<u>XGBoost</u>	27.2594	5564.3393	74.5945	0.8069

Results

3. System / Application Results

Land/property data was prepared for machine learning-based price prediction.

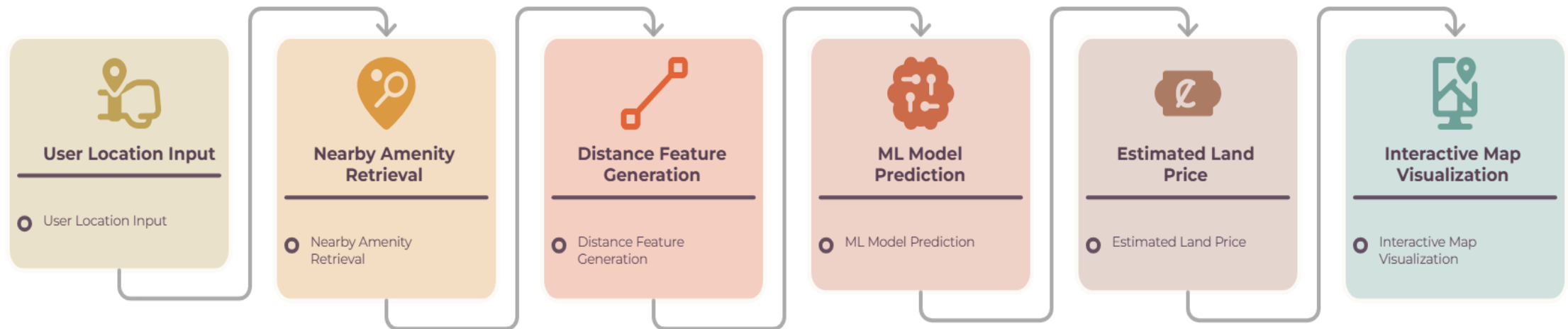
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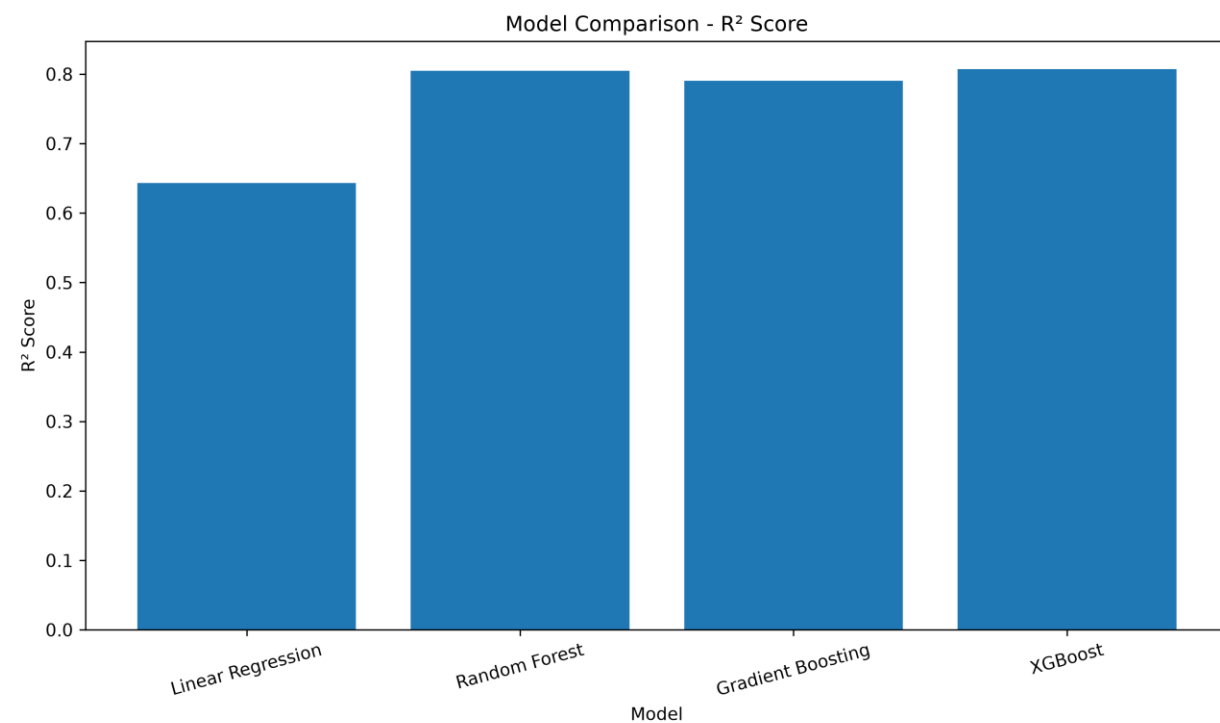
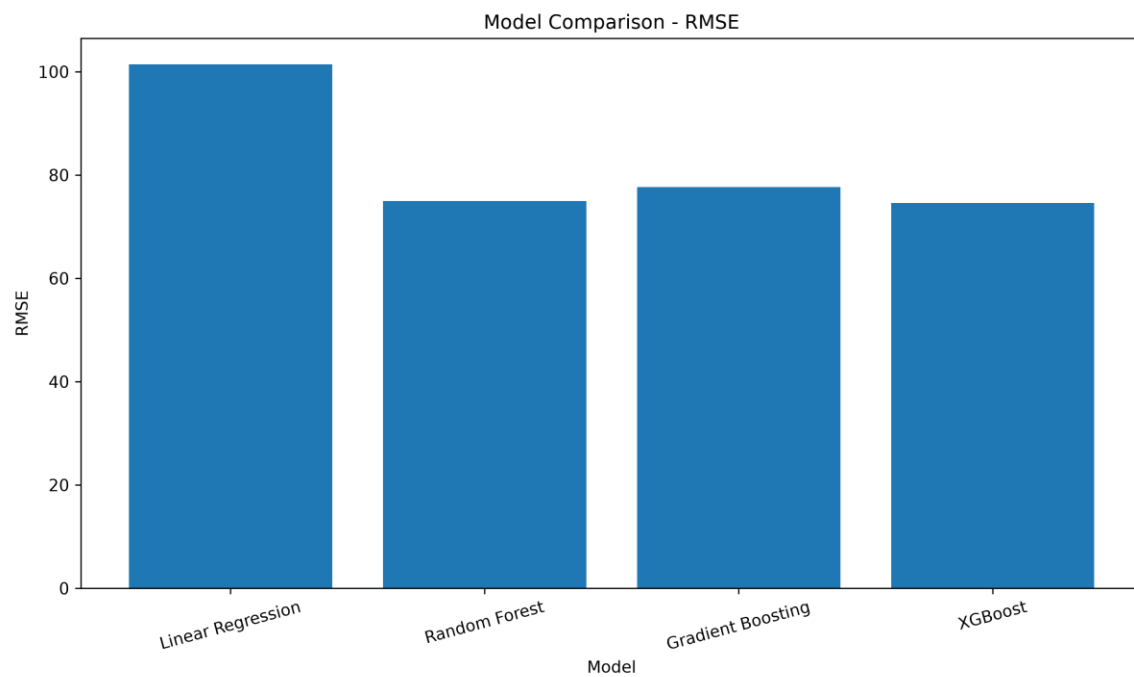
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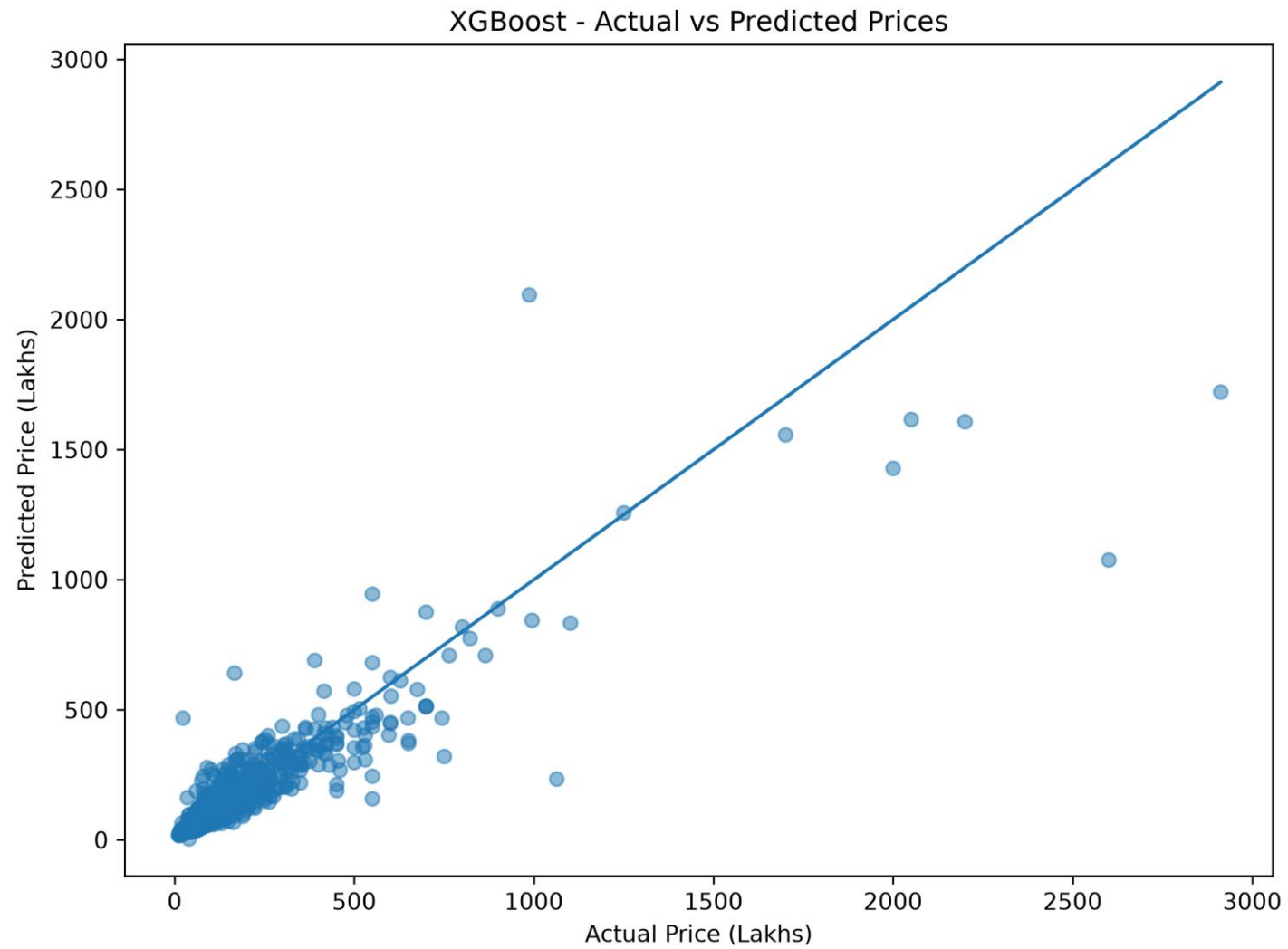
Land Price Estimation Process



Results



Results



References

- [1] A. Soltani and C. L. Lee, “The non-linear dynamics of South Australian regional housing markets: A machine learning approach,” *Applied Geography*, vol. 166, Art. no. 103248, 2024, doi: 10.1016/j.apgeog.2024.103248.
- [2] H. Wei, Y. Chen, B. Chen, S. Shi, Y. Tu, and B. Xu, “Built environments, communities, and housing price: A data-model integration approach,” *Applied Geography*, vol. 166, Art. no. 103270, 2024, doi: 10.1016/j.apgeog.2024.103270.
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- [4] H. Lee, H. Han, and C. Pettit, “Capturing the distance decay effect of amenities on housing price using explainable artificial intelligence,” *Applied Geography*, vol. 174, Art. no. 103486, 2025, doi: 10.1016/j.apgeog.2024.103486.
- [5] P. Zhang, S. Yang, J. Huang, and S. Hu, “Block-scale modeling of residential land prices: Incorporating multilevel determinants and explainable artificial intelligence,” *Habitat International*, vol. 156, Art. no. 103283, 2025, doi: 10.1016/j.habitatint.2024.103283.
- [6] P. Jafary, D. Shojaei, A. Rajabifard, and T. Ngo, “Automated land valuation models: A comparative study of four machine learning and deep learning methods based on a comprehensive range of influential factors,” *Cities*, vol. 151, Art. no. 105115, 2024, doi: 10.1016/j.cities.2024.105115.

Thank You!